

[2] Given:  $\dot{m} = 0.1 \text{ kg/s}$

$$P_1 = P_4 = 0.16 \text{ MPa}$$

$$P_2 = P_3 = 1.8 \text{ MPa}$$

$$x_1 = 1$$

$$x_3 = 0$$

$$\eta_{\text{comp}} = 0.9$$

Assume: S.C.S.  $+1/2$

: quasi-g.  $+1/2$

: steady-flow  $+1/2$

$+1/2$ : compressor:  $\dot{m}_{\text{ke}} = \dot{m}_{\text{pe}} = \dot{Q} = 0$

$+1/2$ : condenser/evaporator:

$$\dot{m}_{\text{ke}} = \dot{m}_{\text{pe}} = \dot{Q} = 0$$

$+1/2$ : throttle:  $\dot{m}_{\text{ke}} = \dot{m}_{\text{pe}} = \dot{Q} = \dot{W} = 0$

a) Cycle powered by compressor

$$\rightarrow \text{Steady-flow: } \dot{m} (\Delta h + \Delta \cancel{pe} + \Delta \cancel{pe}) = \cancel{\dot{Q}} - \dot{W}$$

$$\rightarrow \text{compressor: } \dot{m} (h_2 - h_1) = -\dot{W} \quad +2$$

$$\dot{W} = -\dot{m} (h_2 - h_1)$$

$$\rightarrow \text{State 1: } P_1 = 0.16 \text{ MPa}$$

$$x_1 = 1$$

$$h_1 = h_g = 389.27 \text{ kJ/kg} \quad +1$$

$$s_1 = s_g = 1.7376 \text{ kJ/kg}\cdot\text{K}$$

$$\rightarrow \text{State 2s: } P_2 = 1.8 \text{ MPa}$$

$$s_{2s} = s_1 = 1.7376 \text{ kJ/kg}\cdot\text{K} \rightarrow s_{2s} > s_g \rightarrow \text{vapor}$$

$$h_{2s} = 439.93 \text{ kJ/kg} \quad +1$$

$$\rightarrow \text{State 2: } \eta_{\text{comp}} = \frac{\dot{W}_{\text{ideal}}}{\dot{W}_{\text{real}}} = \frac{h_{2s} - h_1}{h_2 - h_1} \quad +2$$

$$h_2 = h_1 + \frac{h_{2s} - h_1}{\eta_{\text{comp}}} = 389.27 + \frac{439.93 - 389.27}{0.9}$$

$$h_2 = 445.56 \text{ kJ/kg} \quad +1$$

$$\Rightarrow \dot{W} = - (0.1) (445.56 - 389.27)$$

$$\boxed{\dot{W} = -5.629 \text{ kW}} \quad +1$$

b) Remove from cold-space  $\rightarrow$  evaporator

$$\rightarrow \text{Steady-flow: } \dot{m}(\Delta h + \Delta e + \Delta p e) = \dot{Q} - \dot{W}$$

$$\rightarrow \text{Evaporator: } \dot{m}(h_1 - h_4) = \dot{Q} + 2$$

$$\rightarrow \text{State 4: } h_4 = h_3 = 292.26 \frac{\text{kJ}}{\text{kg}} \leftarrow +1$$

$$\rightarrow \text{State 3: } P_3 = 1.8 \text{ MPa} \quad h_3 = h_f = 292.26 \frac{\text{kJ}}{\text{kg}} +1$$

$$x_3 = 0 \quad s_3 = s_f = 1.2987 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

$$\Rightarrow \dot{Q} = \dot{m}(h_1 - h_4)$$

$$= (0.1)(389.27 - 292.26)$$

$$\boxed{\dot{Q} = 9.701 \text{ kW}} +1$$

$$c) \beta_{\text{frig}} = \frac{\dot{Q}_H}{\dot{W}_{\text{in}}} = \frac{9.701}{5.629} \rightarrow \boxed{\beta_{\text{frig}} = 1.72} +1$$

$$\beta_{\text{Carnot}} = \frac{T_L}{T_H - T_L} = \frac{T_4}{T_{2s} - T_4} = \frac{257.56}{345.36 - 257.56}$$

$$\boxed{\beta_{\text{Carnot}} = 2.93} +1$$

$$d) \text{State 4: } s_{4,s} = s_3 = 1.2987 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} \rightarrow s_f < s_{4,s} < s_g$$

$$P_4 = 0.16 \text{ MPa}$$

$$x_{4,s} = \frac{1.2987 - 0.92262}{1.7376 - 0.92262} = 0.46 +1$$

$$h_{4,s} = x_{4,s} h_g + (1 - x_{4,s}) h_f$$

$$= (0.46)(389.27) + (1 - 0.46)(179.37)$$

$$= 275.924 \frac{\text{kJ}}{\text{kg}} +1$$

$$\dot{W} = -\dot{m}(h_{4,s} - h_3) = -0.1(275.924 - 292.26)$$

$$\boxed{\dot{W} = 1.634 \text{ kW}} +2$$

[3] Given:  $P_2/P_1 = 12.2$

:  $\dot{m} = 21.5 \text{ kg/s}$

:  $\dot{Q} = 18 \text{ MW}$

:  $\eta_{\text{comp}} = 0.85$

$\eta_{\text{turb}} = 0.9$

:  $T_1 = 278 \text{ K}$   $P_1 = 95900 \text{ Pa}$

Assume: air standard +1/2

: quasi-eq +1/2

: steady-flow +1/2

+1  $\rightarrow$  compressor:  $\Delta ke = \Delta pe = \dot{Q} = 0$   
turbine

$\rightarrow$  combustor:  $\Delta ke = \Delta pe = \dot{W} = 0$   
+1/2

a) State 1:  $T_1 = 278 \text{ K}$   
 $P_1 = 95900 \text{ Pa}$

State 2:  $P_2 = (PR) P_1^{+1} = (12.2)(95900) \rightarrow P_2 = 1169980 \text{ Pa}$  +1

:  $T_{2s} = T_1 (PR)^{\frac{\gamma}{\gamma-1} +2} = 278 (12.2)^{\frac{0.4}{1.4}} = 568.1 \text{ K}$  +1

:  $\eta_{\text{comp}} = \frac{\dot{W}_{\text{ideal}}}{\dot{W}_{\text{real}}} = \frac{T_{2s} - T_1}{T_2 - T_1}$  +2

$T_2 = T_1 + \frac{T_{2s} - T_1}{\eta_{\text{comp}}} = 278 + \frac{568.1 - 278}{0.85}$

$T_2 = 619.29 \text{ K}$  +1

State 3: steady flow  $\rightarrow \dot{m}(\Delta h + \Delta ke + \Delta pe) = \dot{Q} - \dot{W}$

combustor  $\rightarrow \dot{m}(h_3 - h_2) = \dot{Q}$  +2

ideal gas,  $c_p = \text{const} \rightarrow \dot{m} c_p (T_3 - T_2)^{+1} = \dot{Q}$

$\Rightarrow T_3 = T_2 + \frac{\dot{Q}}{\dot{m} c_p} = 619.29 + \frac{18 \times 10^6}{(21.5)(1001)}$

$T_3 = 1455.7 \text{ K}$  +1

$P_3 = P_2 = 1169980 \text{ Pa}$  +1

State 4s :  $P_4 = P_1 = 95900 \text{ Pa}$  +1

$$T_{4s} = T_3 \left( \frac{P_4}{P_3} \right)^{\frac{\gamma-1}{\gamma}} = 1455.7 \left( \frac{1}{12.2} \right)^{\frac{0.4}{1.4}} = 712.33 \text{ K} +1$$

State 4 :  $\eta_{\text{turb}} = \frac{\dot{W}_{\text{real}}}{\dot{W}_{\text{ideal}}} = \frac{T_4 - T_3}{T_{4s} - T_3}$  +2

$$T_4 = T_3 + \eta_{\text{turb}} (T_{4s} - T_3) = 1455.7 + 0.9(712 - 1455.7)$$

$T_4 = 786.67 \text{ K}$  +1

b)  $\dot{W}_{\text{net}} = \dot{W}_{\text{comp}} + \dot{W}_{\text{turb}}$  +2

$$= -\dot{m} c_p (T_2 - T_1) - \dot{m} c_p (T_4 - T_3)$$

$$= -21.5(1001) [(619.29 - 278) + (786.67 - 1455.7)]$$

$\dot{W}_{\text{net}} = 7.05 \text{ MW}$  +1

c)  $\eta_{\text{th}} = \frac{\dot{W}_{\text{net}}}{\dot{Q}_{\text{in}}} = \frac{7.05}{18} \rightarrow \eta_{\text{th}} = 0.392$  +2

[4] Given:  $T_1 = 300 \text{ K}$

$P_1 = 0.1 \text{ MPa}$

$V_1 = 0.625 \text{ L} = 0.000625 \text{ m}^3$  : reversible  $+1/2$

$zQ_3 = 1500 \text{ J}$

$r = \frac{V_1}{V_2} = 11$   $+1/2$

Assume: air standard  $+1/2$

: quasi-st  $+1/2$

:  $M = \text{const}$   $+1/2$

:  $\Delta KE = \Delta PE = 0$   $+1/2$

a) Ideal gas  $\rightarrow P_1 V_1 = MRT_1 \rightarrow M = \frac{P_1 V_1}{RT_1} = \frac{(100000)(0.625 \times 10^{-3})}{(287)(300)} = 0.000726 \text{ kg}$   $+1$

State 1:

$T_1 = 300 \text{ K}$   
 $P_1 = 0.1 \text{ MPa}$   
 $V_1 = 0.000625 \text{ m}^3$

State 2:  $V_2 = \frac{V_1}{r} = \frac{0.000625}{11} \rightarrow V_2 = 0.0000568 \text{ m}^3$   $+1/2$

$P_2 = P_1 r^{1.4} = 0.1 (11)^{1.4} \rightarrow P_2 = 2.87 \times 10^6 \text{ Pa}$   $+1/2$

$T_2 = \frac{P_2 V_2}{MR} = \frac{(2.87 \times 10^6)(0.0000568)}{(0.000726)(287)} \rightarrow T_2 = 782.4 \text{ K}$   $+1/2$

State 3:

$\Delta U = Q - W$   $+2$   
 $\rightarrow$  where  $\Delta V = 0$

Ideal gas:  $Mc_v(T_3 - T_2) = zQ_3$   $+1$

$T_3 = T_2 + \frac{zQ_3}{Mc_v} = 782.4 + \frac{1500}{(0.000726)(714)}$

$T_3 = 3676.12 \text{ K}$   $+1/2$   
 $V_3 = V_2 = 0.0000568 \text{ m}^3$   $+1/2$

$P_3 = \frac{MRT_3}{V_3} = \frac{(0.000726)(287)(3676.12)}{0.0000568} \rightarrow P_3 = 13.49 \text{ MPa}$   $+1/2$

State 4:  $V_4 = V_1 = 0.000625 \text{ m}^3$   $+1/2$

$$P_4 = P_3 \left(\frac{1}{r}\right)^{\gamma} = 13.49 \left(\frac{1}{11}\right)^{1.4} = 469961.4 \text{ Pa} \quad +1/2$$

$$T_4 = \frac{P_4 V_4}{MR} = \frac{(469961.4)(0.000625)}{(0.000726)(287)} \rightarrow T_4 = 1409.7 \text{ K} \quad +1/2$$

Diesel

State 1:

$$\begin{aligned} T_1 &= 300 \text{ K} \\ P_1 &= 0.1 \text{ MPa} \\ V_1 &= 0.000625 \text{ m}^3 \end{aligned}$$

State 2: Same compression as Otto:

$$\begin{aligned} T_2 &= 782.4 \text{ K} \\ V_2 &= 0.0000568 \text{ m}^3 \\ P_2 &= 2.87 \text{ MPa} \end{aligned} \quad +1/2$$

State 3:  $\Delta U = {}_2Q_3 - {}_2W_3$   $+2$

$$M C_V (T_3 - T_2) = {}_2Q_3 - P(V_3 - V_2)$$

$$\Rightarrow PV = MRT$$

$$M C_V (T_3 - T_2) = {}_2Q_3 - MR(T_3 - T_2) \quad +2$$

$$M \underbrace{(C_V + R)}_{C_P} (T_3 - T_2) = {}_2Q_3 = M C_P (T_3 - T_2)$$

$$T_3 = T_2 + \frac{{}_2Q_3}{M C_P} = 782.4 + \frac{1500}{(0.000726)(1000)}$$

$$T_3 = 2848.52 \text{ K} \quad +1/2$$

$$P_3 = P_2 = 2.87 \text{ MPa} \quad +1/2$$

$$V_3 = \frac{M R T_3}{P_3} = \frac{(0.000726)(287)(2848.52)}{2.87 \times 10^6}$$

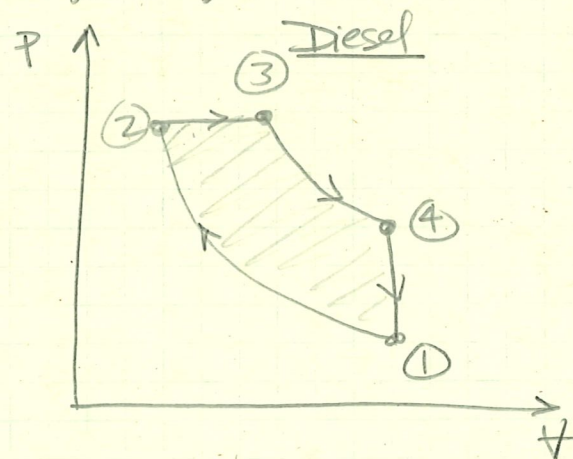
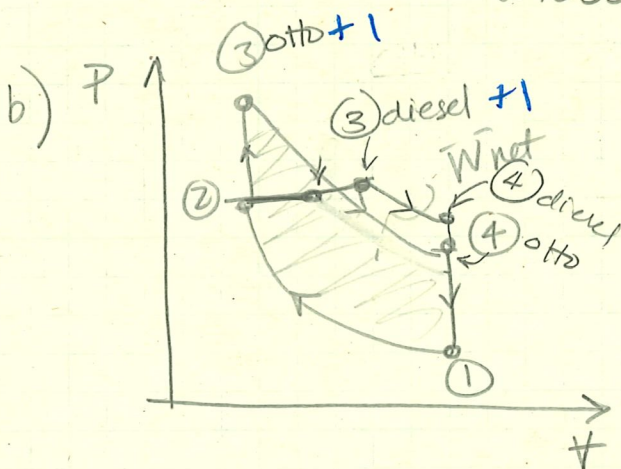
$$V_3 = 0.00021 \text{ m}^3 \quad +1/2$$

State 4:  $V_4 = V_1 = 0.000625 \text{ m}^3$   $+1/2$

$$P_4 = P_3 \left( \frac{V_3}{V_4} \right)^{\gamma} = (2.87 \times 10^6) \left( \frac{0.00021}{0.000625} \right)^{1.4}$$

$P_4 = 623385.77 \text{ Pa}$   $+1/2$

$$T_4 = \frac{P_4 V_4}{MR} = \frac{(623385.77)(0.000625)}{(0.000726)(287)} = T_4 = 1869.9 \text{ K} \quad +1/2$$



c)  $\bar{W}_{net,otto} = {}_1\bar{W}_2 + {}_3\bar{W}_4 = -MCW[(T_2 - T_1) + (T_4 - T_3)]$   $+1$   
 $= (0.000726)(714)[(782.4 - 300) + (1409.7 - 3676.12)]$

$$\bar{W}_{net,otto} = 924.77 \text{ J}$$

$$\eta_{otto} = \frac{924.77}{1800} \rightarrow \eta_{otto} = 0.617 \quad +1$$

$$\bar{W}_{net,diesel} = {}_1\bar{W}_2 + {}_2\bar{W}_3 + {}_3\bar{W}_4 \quad +1$$

$$= -MCW(T_2 - T_1) + P_2(V_3 - V_2) - MCW(T_4 - T_3)$$

$$= -(0.000726)(714)[(782.4 - 300) + (1869.9 - 2848.52)]$$

$$+ (2.87 \times 10^6)(0.00021 - 0.0000568)$$

$$\bar{W}_{net,diesel} = 696.9 \text{ J}$$

$$\eta_{diesel} = \frac{696.9}{1800} \rightarrow \eta_{diesel} = 0.465 \quad +1$$